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題目: Ultra Lightweight Dehaze Methods for Robot Vision Using High-Level Synthesis

FPGA 發展趨勢

技術

— 成本↓
— 雲端AWS導入大量FPGA

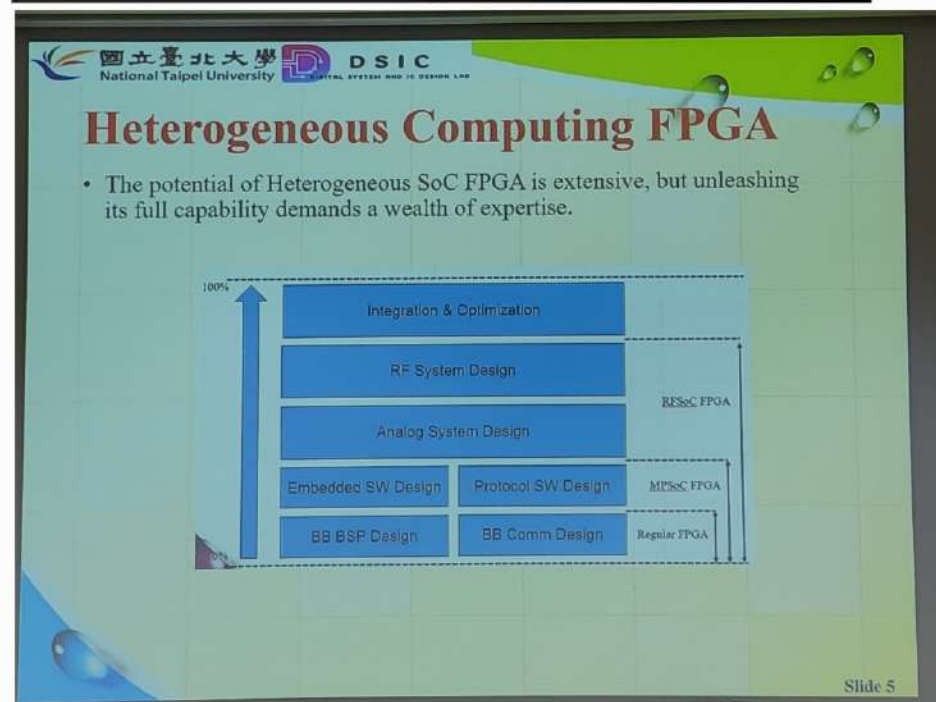
應用

— 異質加速
— SDR (軟體定義無線電)
— AI/Vision

異質運算FPGA

多層次異質系統整合

Baseband
Embedded Software
Protocol Software
Analog / RF System
System Integration & Optimization



HLS

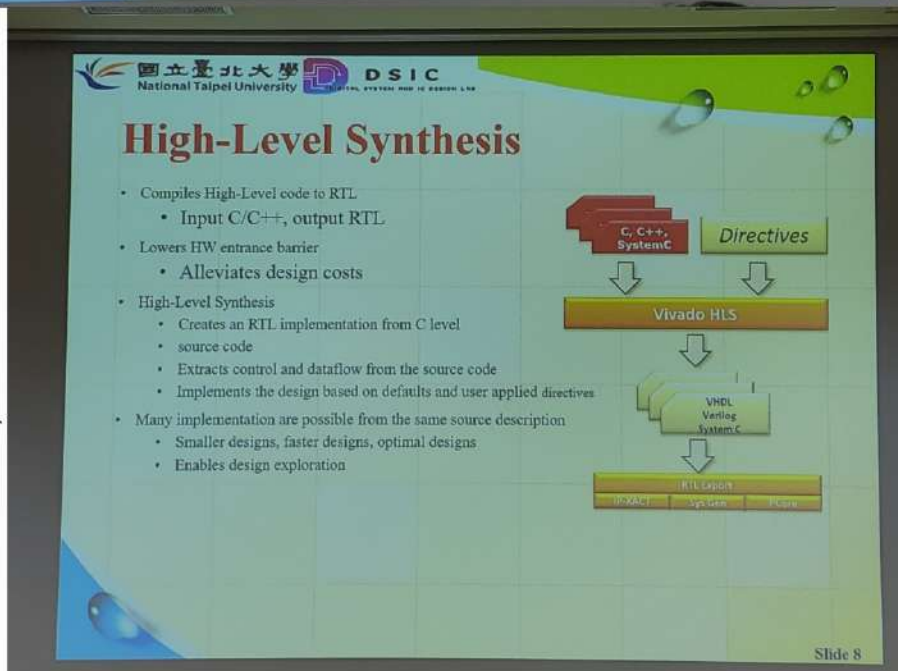
用C/C++/System C

自動產生RTL

— 降低門木檻
— 縮短開發時間

用在矩陣運算量大時,
使用HLS效果會比手寫RTL好
若用在串流接收訊號時則好

PYQ N (Python + FPGA)



案例使用HLS比寫RTL
在資源使用上佳
開發時間短
實作效果延遲低

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HLS (Autopilot) Results: Sphere Decoder

- Wireless MIMO Sphere Decoder
 - ~4000 lines of C code
 - Xilinx Virtex-5 at 225MHz
- Compared to optimized IP
 - 11-31% better resource usage

TCAD April 2011 (keynote paper)
"High-Level Synthesis for FPGAs: From Prototyping to Deployment"

Metric	RTL Expert	AutoPilot Expert	Diff (%)
LUTs	32,708	29,060	-11%
Registers	44,885	31,000	-31%
DSP48s	225	201	-11%
BRAMs	128	99	-26%

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HLS (Autopilot) Results: QRD

2013 Xilinx All Programmable show that Vivado High Level Synthesis can achieve "For each project where we used Vivado HLS, we saved 2-3 weeks of engineering time."

Radar Design 1024 x 64 QRD Floating Point data path	Conventional Hand-coded HDL Approach	Using Vivado High Level Synthesis
Design Language	VHDL (RTL)	C
Design Time (weeks)	12	1
Latency (ms)	37	21
Memory (RAMB18E1)	134 (16%)	10 (1%)
Memory (RAMB36E1)	273 (65%)	138 (33%)
Registers	29686 (9%)	14263 (4%)
LUTs	28152 (18%)	24257 (16%)

CTO, Major broadcast equipment company "2013 Xilinx All Programmable"

左圖面積小速度較慢
右圖面積大速度較快

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Optimization I: Loop unrolling

- Exploits loop iteration parallelism
 - Instantiate resources for simultaneous execution
 - Respects iteration dependencies

Standard HLS

Takes 4 Cycles

HLS & Unrolling

Takes 1 Cycles

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HLS把Array實作成RAM

一次只能取有限個Port

1. block

將array切成連續區塊，
適合區段式存取

2. cyclic

適合unrolling loop 同時存取不同index

3. complete

每個element變成一個register

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Optimization II: Array partitioning

- Split arrays to improve memory bandwidth

array[N]

block: Divided into blocks: $N/2$ factor elements

cyclic: Divided into blocks: 1 word at a time (like "dealing cards")

complete: Individual elements: Break a RAM into registers (no "factor" supported)

Multiple memories allows greater parallel access

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限制硬體資源數量(省LUT/DSP)

限制加法器乘法器

指定乘法器型態

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Optimization III: Resources

- Allocation directive constrains resources
 - Operations
 - Number of adders instantiated RTL
 - Can save a lot of area
- Specify implementation
 - Tag operator
 - Select core for *thisMult*

```
thisMult = b[i] * c[i];  
a[i] = thisMult;
```

Core	Description
Mul	Combinational mult
Mul3s	3-Stage pipelined mult
MulnS	HLS determine stages

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memory/port不夠需要提升throughput

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Optimization IV: Loop Pipelining

- Iteration Interval (II)
 - Cycles loop must before next iteration
- II = 1 cannot be implemented
 - Port cannot be read at the same time
 - Similar effect with other resource limitations

(A) Pipeline with II=1

II=1

```
void foo(m[2]...) {  
  op_Read_m[0];  
  op_Read_m[1];  
  op_Compute;  
  op_Write;  
}
```

(B) Pipeline with II=2

II=2

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雨霧降低 Robot Vision
影響 YOLO 等模型

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Lightweight Dehazer Circuit System Design for Robot Vision

- Naturally, fog causes degraded vision in imaging
- Degrades performance of Robot Vision using Machine Learning, like yolo or ViT models
 - Object detection
 - Classification
 - Tracking
 - Segmentation



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降低計算量 提升速度

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Main Objective

- Develop a ultra dehazing (defog, derain, low light, under water) algorithm that performs better or on-par with current works
- Eliminate or minimize the drawbacks of existing method
- Minimizing computation requirements
- Implementation in FPGA as an IP using HLS
- Integrate into HDMI pipelining

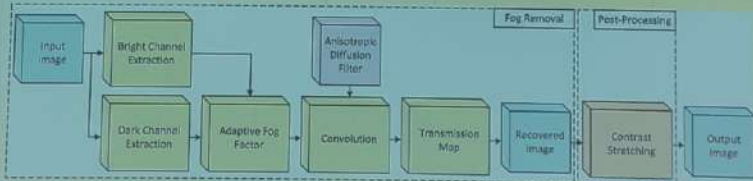
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同時使用 Dark + Bright channel
加入 Adaptive Fog Factor
減少色偏

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Dehaze Method

- Dark channel extraction
- Bright channel extraction
- Transmission map estimation
- Scene recovery



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消除DCP偏藍偏綠
天空、建築顏色自然

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Dehazing Results

- Eliminates color shifting from original DCP
- Preserves image naturalness

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PSNR 不高(超過30)
對 Robot Vision 綽綽有餘
FPS 因計算量大符降低

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Dehazing Metrics on RESIDE

- Tested on MATLAB R2024b using CPU-only process
- High PSNR and SSIM compared to conventional method
- Higher throughput compared to other methods

Benchmark metrics comparison on RESIDE dataset

Metrics	DCP [4]	BCCR [14]	CAP [8]	NLD [15]	Dehaze-Net [17]	Dehaze-Former [21]	MAPD [16]	Proposed
FPS ↑	0.23	4.39	12.27	3.97	2.73	0.42	2.46	384.17
PSNR ↑	17.475	14.955	18.221	18.239	22.917	37.218	22.006	22.101
SSIM ↑	0.821	0.768	0.785	0.784	0.888	0.989	0.881	0.898
BRISQUE ↓	20.695	17.760	21.468	20.879	21.287	20.099	25.234	19.262
NIQE ↓	2.686	2.750	2.778	2.816	2.681	2.437	2.757	2.709
ILNIQE ↓	21.625	20.526	23.853	21.383	22.181	20.721	21.754	22.139
BIQI ↓	22.278	22.091	21.364	22.799	21.586	21.239	27.316	21.057
IDME ↓	0.205	0.220	0.202	0.218	0.203	0.201	0.191	0.208
RMSE ↓	35.869	45.317	32.649	32.233	19.190	3.241	20.994	21.769
EMEE ↓	24.656	45.676	37.602	33.529	16.803	3.257	20.677	20.897
AME ↓	34.965	47.972	36.760	37.409	20.613	4.515	27.063	23.501

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